

AP Biology Summer Assignment – 2026/27

Mr. Smitka – A147 – rsmitka@roseville.k12.mi.us

Welcome to AP Biology! This course is designed to be similar to a two-semester introductory college biology course. AP Biology emphasizes scientific reasoning, data analysis, experimental design, and explanations supported by evidence. This summer assignment is designed to help you begin thinking like a biologist before the school year starts.

Important: All assignments are due on the **second day** of school. Do not wait until the last week of summer to begin. If technology access is an issue, contact me before summer break or as soon as possible at rsmitka@roseville.k12.mi.us

The Four Big Ideas of AP Biology

- Big Idea 1: Evolution drives the diversity and unity of life.
- Big Idea 2: Biological systems use energy and molecular building blocks to grow, reproduce, and maintain homeostasis.
- Big Idea 3: Living systems store, retrieve, transmit, and respond to information.
- Big Idea 4: Biological systems interact, and those interactions produce complex properties.

AP Biology Science Practices

Science Practice	How You Will Use It
SP1 - Concept Explanation	Explain biological processes and concepts in your own words.
SP2 - Visual Representations	Create and interpret graphs, tables, and models.
SP3 - Questions and Methods	Ask testable questions and design investigations.
SP4 - Data Analysis	Analyze patterns, trends, and sources of variation in data.
SP5 - Statistical Tests	Use mean, standard deviation, SEM, error bars, and chi-square.
SP6 - Argumentation	Use evidence and reasoning to support biological claims.

Assignment Overview

	Focus	What You Will Do	What You Submit
1	Thinking Like a Biologist	Analyze biological data, interpret graphs, evaluate reliability, and explain scientific reasoning using evidence	Completed calculations, data tables, graphs, and written analysis questions
2	AP Biology Concept Photo Investigation	Observe and explain real-world biology using AP Biology concepts and scientific thinking	Digital portfolio or printed photo pages with original photos and explanations
3	AP Biology Investigation Design	Design a biological investigation by developing a question, identifying variables, planning data collection, and predicting results	A completed experimental investigation design with tables, graphs, and scientific explanations

Assignment #1 – Thinking Like a Biologist

Biologists use data to identify patterns, evaluate reliability, and support scientific conclusions. In AP Biology, you will analyze investigations, interpret graphs, evaluate evidence, and explain biological reasoning.

Investigation 1 – Are These Results Reliable?

Scientists repeat experiments because biological systems naturally vary. In this investigation, you will calculate statistics and determine whether results appear reliable.

Watch this video for help: <https://tinyurl.com/APBio-StDev>

Key AP Biology Terms

Term	Meaning
Mean	The average value of a data set
Standard Deviation	A measure of how spread out data are from the mean
Variability	Natural differences found in biological data

Why Do Biologists Use Standard Deviation?

Biologists use standard deviation to measure how much data vary from the mean. It helps scientists determine whether results are consistent and reliable or whether there is a large amount of variability in the data.

Standard Deviation Formula

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Σ = Summation $\sqrt{}$ = Square root n = Number of points of data x = data point $\bar{x}$ = mean

You are NOT expected to memorize this formula. Focus on understanding what the calculations tell scientists about the data.

Worked Example

A student measures the height of three plants.

Plant Height (x)	x - mean	(x - mean) ²
5	-1	1
6	0	0
7	1	1

Mean = (5 + 6 + 7) ÷ 3 = 6

Seed Germination Investigation

A biology class measured the number of seeds that germinated in five identical trays.

Tray	Seeds Germinated (x)	x - mean	(x - mean) ²
1	8		
2	4		
3	6		
4	6		
5	7		
Totals			

Step-by-Step Calculations

Step	Student Work
1. Add all data values together	
2. Divide by the number of data points to calculate the mean	
3. Subtract the mean from each data point	
4. Square each difference	
5. Add all squared differences together	
6. Calculate the standard deviation	

Analysis Questions

1. Describe the variability in this data set using evidence from the data.

2. Explain why variability is expected when working with living systems.

3. Explain how replication and larger sample sizes improve scientific reliability.

Investigation 2 – Do These Groups Appear Different?

Scientists compare groups to determine whether differences appear meaningful or could simply be due to natural variation.

Watch this video for help: <https://tinyurl.com/APBio-StError>

Key AP Biology Terms

Term	Meaning
SEM	A measure of how precisely a sample estimates the true population mean
Error Bars	Lines on a graph showing variability or uncertainty

Why Do Biologists Use SEM and Error Bars?

The Standard Error of the Mean (SEM) measures how precisely a sample mean estimates the true population mean. Biologists use SEM and error bars to evaluate the reliability of data and compare groups visually. Smaller error bars usually suggest more consistent data, while overlapping error bars may indicate that differences between groups are not biologically significant.

Sunflower Stomata Investigation

Group	1	2	3	4	5	6
Group A	88	93	90	92	75	78
Group B	67	85	90	100	72	97

Group	n	Mean	Standard Deviation	SEM	+/- 2 SEM
Group A					
Group B					

Step-by-Step Calculations

Step	Student Work
1. Calculate the mean for each group	
2. Calculate the standard deviation for each group	
3. Calculate SEM for each group	
4. Calculate ± 2 SEM	
5. Graph the data with error bars	

Create a bar graph showing the mean stomatal density for Group A and Group B with ± 2 SEM error bars.

Graph Analysis Questions

1. Which group has the higher mean stomatal density?

2. Do the error bars overlap?

3. What might the overlap (or lack of overlap) suggest about the groups?

4. Identify one limitation of this investigation.

5. Suggest one improvement scientists could make.

Investigation 3 – Was This Experiment Designed Well?

Strong investigations control variables carefully so scientists can identify cause-and-effect relationships.

Scenario 1 – Fertilizer and Plant Growth

A student investigates how fertilizer concentration affects bean plant growth.

Component	Student Response
Independent Variable	
Dependent Variable	
Controlled Variable #1	
Controlled Variable #2	
Control Group	
Why is the control group important?	

Scenario 2 – Temperature and Enzyme Activity

A student investigates how temperature affects enzyme activity.

Component	Student Response
Independent Variable	
Dependent Variable	
Controlled Variable #1	
Controlled Variable #2	
Control Group	
Why is the control group important?	

Investigation 4 – Did the Results Match the Prediction?

Scientists compare observed data to expected results to determine whether differences are likely due to chance.

Watch this video for help: <https://tinyurl.com/APBio-Null-Chi>

Why Do Biologists Use Null Hypotheses?

Biologists use null hypotheses as a starting point for statistical analysis. A null hypothesis assumes that there is no significant difference between the observed data and the expected data, allowing scientists to test whether experimental results are likely due to chance.

Example

Question: Does fertilizer type affect tomato plant growth?

Null Hypothesis:

- There is no significant difference in tomato plant growth among the different fertilizer treatments.

Another Example

Question: Does temperature affect bacterial colony growth?

Null Hypothesis:

- There is no significant difference in bacterial colony growth at the different temperatures tested.

Null Hypothesis Practice

1. Do plants grow taller under blue light than under red light?

Null Hypothesis:

2. Does caffeine increase heart rate?

Null Hypothesis:

3. Does salt concentration affect osmosis in potato cells?

Null Hypothesis:

Genetics and Chi-Square Investigation

Chi-Square Formula

Chi-square Formula

$$\chi^2 = \sum \frac{(o - e)^2}{e}$$

χ^2 = Chi Square

Σ = Sum

O = Observed
data
(collected data)

E = Expected
values
(calculated
data)

Chi-Square Table

<i>p</i> value	Degrees of Freedom							
	1	2	3	4	5	6	7	8
0.05	3.84	5.99	7.81	9.49	11.07	12.59	14.07	15.51
0.01	6.63	9.21	11.34	13.28	15.09	16.81	18.48	20.09

Why Do Biologists Use Chi-Square Analysis?

Biologists use chi-square analysis to determine whether differences between observed data and expected data are likely due to chance or whether the differences may be biologically meaningful. Chi-square is commonly used in genetics to compare observed offspring ratios to predicted Mendelian ratios.

In a genetics cross, the expected ratio is 3 purple-flowered plants : 1 white-flowered plants

A student observed:

- 56 purple-flowered plants
- 24 white-flowered plants

Phenotype	Observed (O)	Expected Ratio	Expected (E)	O - E	$(O - E)^2$	$(O - E)^2 / E$
Purple Flowers	56	3/4				
White Flowers	24	1/4				
Total Chi-Square						

Step-by-Step Calculations

Step	Student Work
1. Calculate the expected number for each phenotype	
2. Subtract expected values from observed values	
3. Square each difference	
4. Divide each squared difference by the expected value	
5. Add all values to calculate chi-square	
6. Determine degrees of freedom	
7. Compare your chi-square value to the critical value table	
8. Decide whether to reject or fail to reject the null hypothesis	

Analysis Questions

1. Write a null hypothesis for this investigation.

2. Do the observed results appear close to the expected ratio? Explain your reasoning.

For the above question, here's what strong AP Biology thinking looks like

Weak Response	Strong AP Biology Response
The data are different.	The data show moderate variability because several values differ noticeably from the mean.
The experiment worked.	The investigation appears reliable because the results show relatively low variability across trials.
The groups are different.	The lack of overlapping error bars may suggest the groups differ significantly.

3. Why might biological data differ slightly from expected ratios?

4. Why is chi-square useful in genetics investigations?

Assignment #2 - AP Biology Concept Photo Investigation

Purpose: Biology is not just vocabulary. Biology is the study of living systems, interactions, structures, functions, and evidence. This assignment asks you to observe biology in the real world and explain what you see using AP Biology concepts.

Directions

- Create a digital portfolio using Google Sites, Canva, Google Slides, PowerPoint, Wix, Weebly, or another free web-publishing format.
- Include 15 original photos. Each photo must represent a different AP Biology concept.
- Each photo must include a small proof object, such as a keychain, pen, bracelet, or small toy, to verify that the photo is your own.
- Do not use images from the internet, textbooks, AI image generators, or other publications.
- Be respectful of living things. Do not harm, remove, or disturb organisms or habitats.
- Humans may be used in a few photos, but most photos should come from natural or biological examples around you.
- Email the website link or completed digital portfolio to rsmitka@roseville.k12.mi.us

Each Photo Entry Must Include

Required Component	What to Write
1. Biological Concept	Name the AP Biology concept shown in the photo.
2. Definition	Define the concept in your own words.
3. Biological Explanation	Explain how the photo demonstrates the concept. Focus on process, mechanism, interaction, or structure and function.
4. Big Idea Connection	Identify BI1, BI2, BI3, or BI4 and explain the connection.
5. AP Unit Connection	Identify the AP Biology unit most closely connected to the concept.
7. Testable Question	Write one scientific question inspired by the photo that could be investigated.

Model Photo Entry



Entry Part	Example
Concept	Detritivore
Definition	A detritivore is an organism that eats dead stuff and helps break it down in the environment.
Biological Explanation	Earthworms break down dead organic matter and recycle nutrients back into the soil.
Big Idea Connection	Big Idea 4: Organisms interact with their environment and help maintain ecosystem function.
AP Unit Connection	Unit 8: Ecology
Testable Question	Does the number of earthworms affect the rate of decomposition in soil?

Photo Entry Template

Required Item	Student Response
Biological Concept	
Definition in Your Own Words	
Biological Explanation	
Big Idea Connection	
AP Unit Connection	
Testable Question	

Suggested Concepts by AP Biology Unit

Unit	Major Vocabulary Terms
Unit 1: Chemistry of Life	atom, element, compound, isotope, ion, covalent bond, ionic bond, hydrogen bond, cohesion, adhesion, polarity, pH, acid, base, buffer, monomer, polymer, dehydration synthesis, hydrolysis, carbohydrate, monosaccharide, polysaccharide, lipid, phospholipid, saturated fat, unsaturated fat, protein, amino acid, peptide bond, enzyme, substrate, active site, nucleic acid, DNA, RNA, nucleotide, ATP, macromolecule
Unit 2: Cell Structure and Function	cell theory, prokaryote, eukaryote, organelle, plasma membrane, phospholipid bilayer, selectively permeable membrane, diffusion, osmosis, facilitated diffusion, active transport, endocytosis, exocytosis, tonicity, hypertonic, hypotonic, isotonic, surface area-to-volume ratio, nucleus, ribosome, endoplasmic reticulum, Golgi apparatus, lysosome, vacuole, mitochondrion, chloroplast, cytoskeleton, cilia, flagella, extracellular matrix
Unit 3: Cellular Energetics	metabolism, anabolic pathway, catabolic pathway, activation energy, enzyme inhibition, competitive inhibition, allosteric regulation, photosynthesis, chlorophyll, light-dependent reactions, Calvin cycle, cellular respiration, glycolysis, Krebs cycle, citric acid cycle, electron transport chain, chemiosmosis, ATP synthase, fermentation, aerobic respiration, anaerobic respiration, NADH, FADH ₂ , proton gradient
Unit 4: Cell Communication and Cell Cycle	cell signaling, ligand, receptor, signal transduction pathway, phosphorylation cascade, second messenger, G protein-coupled receptor, tyrosine kinase receptor, feedback loop, positive feedback, negative feedback, homeostasis, mitosis, meiosis, cytokinesis, chromosome, chromatin, sister chromatids, centromere, spindle fibers, cell cycle, interphase, G ₁ phase, S phase, G ₂ phase, cyclin, cyclin-dependent kinase, apoptosis
Unit 5: Heredity	gene, allele, genotype, phenotype, dominant trait, recessive trait, homozygous, heterozygous, Punnett square, monohybrid cross, dihybrid cross, segregation, independent assortment, linked genes, recombination, crossing over, nondisjunction, haploid, diploid, autosome, sex-linked trait, pedigree, variation, meiosis, fertilization, chromosomal mutation
Unit 6: Gene Expression and Regulation	DNA replication, semiconservative replication, helicase, DNA polymerase, RNA polymerase, transcription, translation, codon, anticodon, mRNA, tRNA, rRNA, ribosome, mutation, substitution, insertion, deletion, frameshift mutation, operon, lac operon, gene regulation, epigenetics, transcription factor, enhancer, silencer, alternative splicing, biotechnology, gel electrophoresis, PCR, recombinant DNA, CRISPR, plasmid
Unit 7: Natural Selection	evolution, natural selection, adaptation, fitness, variation, mutation, genetic drift, gene flow, bottleneck effect, founder effect, Hardy-Weinberg equilibrium, speciation, reproductive isolation, convergent evolution, divergent evolution, homologous structures, analogous structures, phylogeny, cladogram, common ancestor, directional selection, stabilizing selection, disruptive selection, artificial selection
Unit 8: Ecology	ecology, ecosystem, community, population, biome, biosphere, niche, habitat, carrying capacity, density-dependent factor, density-independent factor, trophic level, food chain, food web, primary productivity, biomass, symbiosis, mutualism, commensalism, parasitism, competition, succession, biodiversity, carbon cycle, nitrogen cycle, water cycle, energy pyramid, exponential growth, logistic growth, keystone species

Photo Investigation Rubric

Category	Points	Description
Original photos with proof object	15	One point for each original photo that includes the proof object.
Concept identification and definition	15	Concepts are accurate and definitions are in the student's own words.
Biological explanations	15	Explanations connect each photo to biological processes or interactions.
Big Idea and AP Unit connections	15	Entries correctly connect concepts to AP Biology themes and units.
Testable questions	15	Questions are scientific, testable, and related to the photo.
TOTAL	75	

Assignment #3 – AP Biology Investigation Design

Purpose

AP Biology focuses heavily on scientific thinking and experimental design. Scientists ask questions, identify variables, collect evidence, analyze results, and explain conclusions using biological reasoning.

For this assignment, you will design a biological investigation based on a real-world observation. You do **NOT** need to conduct the experiment.

This assignment is designed to help prepare you for AP Biology free-response questions (FRQs), where students are expected to:

- analyze experiments
- identify variables and controls
- predict results
- explain biological reasoning
- evaluate limitations

Step 1 – Choose a Biological Observation

Choose ONE biological phenomenon that interests you.

You may use:

- one of your photos from Assignment #2
- an observation from nature
- an everyday biological process

Possible Topics:

- plant growth
- enzyme activity
- fermentation
- photosynthesis
- osmosis
- decomposition
- transpiration
- bacterial growth
- biodiversity

Choose a topic that can be tested and measured.

Step 2 – Create Your Investigation Design

In Microsoft Word or Google Docs, type your responses using the headings below. Your completed assignment should be approximately 1–2 pages plus a data table. Use complete sentences and scientific vocabulary whenever appropriate.

1. Research Question

Write ONE clear, testable question.

A strong research question:

- tests the effect of one variable on another variable
- can be measured with data
- is specific and focused

Strong Example:

How does temperature affect the rate of yeast respiration?

Weak Example:

Why is yeast important?

2. Background Information

Write ONE paragraph explaining:

- the biological concept involved
- why the topic matters
- what biology predicts may happen

Tip: Pretend you are explaining the science behind your experiment to another AP Biology student.

3. Hypothesis

Write an if-then-because hypothesis.

Formula:

If _____, then _____ because _____.

Strong Example:

If temperature increases, then yeast respiration rate will increase because enzyme activity generally increases with temperature up to an optimal range.

4. Variables and Control

Identify the following:

- Independent Variable
- Dependent Variable
- At least THREE controlled variables
- Control group

Then explain why the control group is important.

Example:

If testing fertilizer concentration:

- Independent Variable = fertilizer concentration
- Dependent Variable = plant height
- Controlled Variables = water amount, light exposure, soil type
- Control Group = plants with no fertilizer

5. Experimental Procedure

Write a numbered procedure detailed enough that another student could repeat the investigation.

Your procedure should include:

- how the investigation would be set up
- how many samples or trials would be used
- how measurements would be collected
- how often data would be recorded

Tip: A strong procedure is clear, organized, and repeatable.

6. Data Table and Graph Plan

Part A – Blank Data Table

Create ONE blank data table that could be used during the investigation.

Your table should:

- include labels
- include units when appropriate
- organize data clearly

Part B – Graph Plan

Describe:

- what type of graph you would create
- what belongs on the x-axis
- what belongs on the y-axis

Tip: The independent variable usually belongs on the x-axis. The dependent variable usually belongs on the y-axis.

7. Predicted Results

Explain what you think would happen and WHY.

A strong response should:

- make a prediction
- describe expected results or trends
- explain the biology behind the prediction

Example:

Plants exposed to blue light may show greater growth because chlorophyll absorbs blue wavelengths efficiently for photosynthesis.

8. Limitations and Sources of Error

Identify at least TWO possible limitations or sources of error.

Then explain:

- how each could affect the results
- why scientists try to reduce error in experiments

Examples:

- inconsistent measurements
- small sample size
- temperature changes
- human error
- contamination

9. AP Biology Connection

Identify:

- ONE AP Biology Big Idea
- ONE AP Biology Science Practice

Then briefly explain how your investigation connects to each one.

- scientific thinking.

Investigation Design Rubric

Component	Points	What Strong Work Includes
Research Question	5	Question is clear and testable
Background Information	10	Accurate biology and thoughtful explanation
Hypothesis	5	If-then-because format with reasoning
Variables and Control	15	Variables and controls are accurate
Procedure and Replication	15	Procedure is organized and repeatable
Data Table and Graph Plan	10	Table and graph match the investigation
Predicted Results	10	Prediction supported with biological reasoning
Limitations and Error Analysis	10	Realistic limitations explained clearly
AP Biology Connection	5	Correct Big Idea and Science Practice
Scientific Communication	15	Writing is organized and scientific

Total: 100 Points

Expectations For This Summer Assignment

- In this assignment, you may use:
 - calculators
 - spreadsheets
 - support videos
 - online tools to help understand concepts
- You may NOT:
 - submit AI-generated writing as your own
 - copy explanations from websites or AI tools
 - use internet images in the photo investigation
- The goal is to practice your own